

- Centralized notification system covering all modules (matches, messages, hangouts, events, bookings, community activity)
- Privacy & safety controls: block/report users, visibility settings, admin moderation queue
- Role-based access: student, verified tutor, verified volunteer, community admin, platform admin
- Verification workflow (college email and/or admin approval) reused across tutoring, volunteering, and community creation

5. Suggested Tech Stack

- Frontend: React + Tailwind CSS + Lucide icons
- Backend: Node.js
- Database and auth: Firebase Firestore
- Real-time messaging: Socket.io or Firebase Realtime Database
- Authentication: Firebase Auth or JWT-based auth with college email verification
- Hosting: Vercel/Netlify (frontend), Render/Railway (backend)

6. Feature Priority Matrix

All modules below are confirmed for the finalized build. Complexity is a rough guide for sprint/time planning during the hackathon.

Feature Module	Priority	Complexity
User Profiles & Onboarding	Must-Have	Low
Intelligent Skill/Interest Matching	Must-Have	Medium
Interest-Based Groups	Must-Have	Medium
Skill-Exchange Marketplace	Must-Have	Medium
Activity Discovery	Must-Have	Low
Messaging & Collaboration	Must-Have	Medium
Hangout Creator	Must-Have	Low
Campus Events (Movies/Karaoke/Jamming/Sports)	Must-Have	Medium
Car & Auto Pooling	Must-Have	Medium
Errand / Shopping Assistance	Must-Have	Low
Accessibility Volunteering (Scribes, etc.)	Must-Have	Medium
Project / Hackathon Team Finder	Must-Have	Medium
Communities (with polls, admin approval)	Must-Have	High
Peer Tutoring Marketplace	Must-Have	High
Notifications System	Must-Have	Low
Search (global)	Must-Have	Low
Admin / Moderation Panel	Should-Have	Medium
Reporting & Safety Tools	Should-Have	Low

3.5 Accessibility Volunteering (Scribes & Support for Specially-Abled Students)

Connects specially-abled students with peer volunteers for academic support such as exam scribing.

- Volunteer sign-up with course/subject qualification and availability
- Request form for a student needing support, specifying course, exam date, and requirement type (scribe, note-taking, mobility assistance, etc.)
- Matching based on course/subject overlap and availability
- Verification step (e.g., admin or faculty approval) for volunteers before they can be matched, given the sensitivity of this feature
- Confirmation and reminder notifications for both parties

3.6 Project / Hackathon Team Finder

A focused matching module for finding collaborators for hackathons, projects, and competitions.

- Post a team listing: project/hackathon name, required skills/roles, team size, deadline
- Browse open listings, filterable by skill/role needed
- Apply-to-join flow with a short pitch/message from the applicant
- Team owner can accept/reject applicants and mark the listing as filled/closed
- Optional: link to the individual skill profiles built in Section 2.1 to speed up screening

3.7 Communities

Larger, topic-based communities (similar to WhatsApp Communities) sitting above individual interest groups.

- Community feed with posts, comments, and reactions
- Polls within a community (single/multiple choice, with results view)
- Community sub-groups or channels for focused discussion (optional, if time permits)
- "Request a Community" flow: students submit a request (name, purpose, category) which goes to an admin approval queue
- Admin dashboard to review, approve, or reject community requests
- Community membership management (join, leave, member roles such as admin/moderator/member)

3.8 Peer Tutoring Marketplace (Verified Senior Tutors)

Paid, verified peer-to-peer tutoring where senior students teach skills to juniors.

- Tutor application/verification flow for senior students (verification by admin, could include proof of skill/grades/certificates)
- Tutor profile: bio, skills taught, hourly rate, available time slots
- Ratings and review system for tutors, built from past session feedback
- Session booking system against the tutor's available slots
- Booking confirmation and reminder notifications
- Session history for both tutor and student
- Note: actual payment gateway integration is optional for the hackathon demo — a mock/placeholder payment step is acceptable unless the team wants to fully integrate one

4. Cross-Cutting & Non-Functional Requirements

- Fully responsive UI (mobile-first, since students will primarily use phones)
- Global search across users, groups, activities, hangouts, events, and communities
- Unified home dashboard/feed surfacing matches, upcoming hangouts/events, and community activity

- RSVP / join functionality with visible attendee count
- Filter activities by interest tag

2.6 Messaging & Collaboration

- One-to-one real-time (or near real-time) chat between connected students
- Group chat tied to each interest group
- Notification system for new matches, messages, and group activity

3. Extended Features (Team-Proposed Ideas — Finalized for Build)

The following modules were proposed by the team to extend the platform beyond the base requirement. All are confirmed as part of the finalized scope.

3.1 Hangout Creator

Lets any student spin up a casual, ad-hoc hangout for others to join.

- Create a hangout with venue, activity type, date/time, and member limit
- Public or invite-only hangout visibility
- Join request / instant-join based on creator's setting
- Auto-close once the member limit is reached
- List/feed of upcoming hangouts, filterable by activity type and date

3.2 Campus Events (Movies, Karaoke, Jamming, Sports)

A more structured, recurring version of activity discovery for organized/larger events.

- Event creation with category (movie night, karaoke, jamming session, sports match, etc.), venue, date/time, and capacity
- Event listing page with category filters and calendar view
- RSVP with waitlist support once capacity is hit
- Option for event organizers (students or clubs) to post updates/announcements to attendees

3.3 Car & Auto Pooling

Helps students coordinate shared rides for common routes (e.g., station, city trips, airport runs).

- Post a ride: source, destination, date/time, vehicle type, and available seats
- Search/browse available rides by route and date
- Request-to-join flow with driver approval
- Basic in-app chat/contact exchange once a ride is confirmed
- Ride history for reference

3.4 Errand & Shopping Assistance ("Bring It For Me")

Students traveling to nearby cities (Vijayawada, Mangalagiri, Guntur) can volunteer to pick up items for others.

- "I'm going to [city] on [date]" post by a volunteering student
- Other students can attach item requests to that trip
- Simple request/accept flow between requester and traveler
- Basic status tracking (requested → accepted → picked up → delivered)
- Optional note field for cost/reimbursement understanding between the two students (platform does not process payments)

Campus Skill & Interest-Based Friend Finder

Finalized Feature Specification Document

1. Purpose & Scope

This document consolidates the mandatory features from the original hackathon problem statement with the additional feature ideas proposed by the team, into a single, finalized feature list. This serves as the baseline scope for development.

2. Core Platform Features (Problem Statement Deliverables)

2.1 User Profiles & Onboarding

Foundation layer for all matching and discovery features

- Sign-up / login, ideally restricted to verified college email addresses
- Profile fields: skills (with proficiency level), interests/hobbies, career goals, department, year of study, availability
- Structured tagging for skills/interests (not free text) so matching logic can operate on them
- Bio and profile photo
- Privacy settings controlling profile visibility and who can message/request connection

2.2 Intelligent Skill & Interest-Based Matching

Core differentiator of the platform

- Matching engine based on shared tags (skills, interests, hobbies, career goals) using a similarity score (e.g. weighted overlap)
- Recommended matches feed with a visible "why you matched" explanation (shared tags)
- Connect / accept / reject request flow
- Filters by department, year, and skill/interest category

2.3 Interest-Based Groups

- Create and join groups organized by category (tech, sports, music, arts, etc.)
- Group discovery page with search and category filters
- Member list and group profile page
- Group-level post/discussion feed

2.4 Skill-Exchange Functionality

- Profile tagging for "Can Teach" vs "Wants to Learn" skills
- Browse/match view specifically for skill-swap pairings (reciprocal matches)
- Request/offer flow to initiate an exchange
- Basic session scheduling between the two matched students

2.5 Activity Discovery

- Listing of campus activities/events with title, time, location, and tags
- Ability for any student to create an activity